THE CORE PAPERS

Realization Calculus

Projectors, Transfer and Arithmetic

Komorebi P.

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A mathematical monograph in twelve chapters.

INTRODUCTION

Objects, operators and arithmetic

Realization Calculus studies how mathematical objects are generated from seeds, resolved into packets and related by operators. Projectors specify the components retained in a description. Transfer records their interaction and the passage from one resolution to another.

The first four chapters develop the Projector Resolved Seed Calculus. The argument then moves through multiresolution signals, evolution and memory, quantum dynamics and packet operator algorithms. The final chapters examine congruences, residual seeds, packet geometry and patching support.

Three questions recur. Which objects can be generated from the available seeds? Which relations survive a change of resolution? What additional hypotheses permit a compatible description to be realized in the intended analytic or arithmetic setting? Each construction is read with its own domain and assumptions.

The arithmetic conclusions retain their conditional character. Conjecture 66 in Chapter XI remains open. Chapter XII requires suitable residual and deformation data, patching and full support, a compatible Hecke point over the same coefficient field, classicality and an attached representation. A universal supply of these inputs is not a conclusion of the chapter.

The mathematical text and visual studies form the main sequence. The formal companions and glossary follow, with sources and references collected at the end.

READING MAP

Following the argument

The main sequence

Read Chapters I-XII in order for the development from seed constructions to conditional arithmetic realization. Each chapter opening connects its subject to the surrounding chapters and identifies the mathematical text and visual studies that follow.

Visual studies

The visual material stays with the chapter whose objects it illustrates. Figures and examples retain their stated inputs and ranges. A chapter identifier accompanies a global plate reference, as in Plate VIII.32.

Formal companions

The Lean guides are collected at the back. They identify the statements, hypotheses and source correspondences in their own formal scope. The machine-native treatment of Chapter I appears with its formal companion.

The back matter

Further studies retain their own hypotheses and open questions. The glossary gives short definitions with links to the relevant treatment. Sources and references are grouped by chapter and treatment at the end of the book.

Begin with Chapter I 1

Open the formal companions 1588

Open the glossary 2647

NOTATION AND REFERENCES

Conventions across the book

Local notation

Each chapter specifies its ambient spaces, scalar fields, domains and index sets. A symbol reused elsewhere need not denote the same object. Read every operator with its stated domain and codomain.

Finite and infinite settings

Finite algebra, limits and infinite-dimensional constructions have different assumptions. A formal identity does not by itself extend an operator to a larger space or supply continuity. The passage between settings is made where the text provides it.

Exact and numerical statements

Exact identities and numerical examples retain their stated roles. Numerical illustrations support the specified comparisons within their parameter ranges.

References within the volume

Continuous page numbers and PDF bookmarks navigate this book. Theorem, equation, figure and section numbers remain local to their named treatment unless a chapter prefix is given. Cite the chapter and treatment with the local number.

Citation labels

Bibliographies and source registers are collected at the end of the book. Their citation labels retain the numbering or short names used in the corresponding chapter and treatment. The accompanying electronic material preserves the supporting sources.

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Glossary 2647

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The scope of the companions

A written proof, a formal declaration and a finite computation answer different questions. The companions state their own coverage and the assumptions on which it depends.

Using the glossary

The glossary is a navigation aid. Its definitions are intentionally brief, and its references lead to the fuller statements and hypotheses in the text.

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Formal companions

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Komorebi P., Realization Calculus: Projectors, Transfer and Arithmetic. When citing an individual result, include its chapter, named treatment and local result number.

CHAPTER I

Projector Resolved Seed Calculus I

Typed operator systems, seed rank, and exact packet coordinates

The opening chapter introduces the seed and operator language used throughout the book.

Seeds, operators and projectors provide the linear language of the series. Exact packet coordinates make the generated objects and the maps between them visible.

IN THIS CHAPTER

Mathematical text 2

Formal companion 1588

Sources and references 2666

Scope

The correspondence concerns the stated proof-level constructions. Noncomputable existence results do not provide executable solvers.

The next chapter compares these descriptions across refinement levels and asks which structures survive that passage.

The central idea

Replacing the abstract grammar algebra by its image $\mathcal{A}_X$ removes syntax that acts as zero. This is a faithful reduction of the action. It does not change any orbit, hull, or computation of seed rank.

2.3 Seeds, addressed hulls, and natural presentations

Definition 2.6 (Linear seed system)**.** A linear seed system is a tuple

$$\mathfrak{S} = (X, \mathcal{A}, S, \mathcal{P}, \tau),$$

where X is a unital left module over $\mathcal{A}$, $\mathcal{A}$ is fixed independently of the target element, $S = (s_1, \dots, s_r)$ is a finite distinguished seed family, $\mathcal{P}$ is either empty or a declared packet system, and τ is either the algebraic convention or a specified Hausdorff topology in which the declared operator action is continuous.

Definition 2.7 (Generated hull)**.** For $S \subseteq X$, define

$$\mathrm{Hull}_{\mathcal{A}}(S) := \sum_{s \in S} \mathcal{A}s.$$

In the topological regime define

$$\mathrm{Hull}_{\mathcal{A},\tau}(S) := \overline{\mathrm{Hull}_{\mathcal{A}}(S)}^{\tau}.$$

For a singleton $S = \{s\}$, write $\mathrm{Hull}_{\mathcal{A}}(s)$.

The algebraic hull uses finite sums of addressed seed vectors. In the topological setting, taking its closure may add limits that no finite expression reaches. Thus algebraic generation and dense generation answer different questions, even when they use the same seeds and operators.

Definition 2.8 (Natural presentation)**.** A seed presentation is natural relative to a declared grammar when the primitive operators and relations are fixed independently of the target, the relation system is finite or effectively enumerable, the topology is intrinsic to the realization, no primitive operator directly stores the complete target, and the presentation exposes a reusable invariant, theorem, or computation not encoded by a primitive chosen for the target.

Proposition 2.9 (Algebra generation and module generation are distinct)**.** *Every nonzero unital algebra A, regarded as a left module over itself, has*

$$\mathrm{srank}_A(A) = 1,$$

with cyclic seed 1_A. This conclusion is independent of the number of algebra generators required to present A as an algebra.

CHAPTER II

Projector Resolved Seed Calculus II

Refinement towers, weighted transfer, and integral separation of operator packets

The linear framework now becomes a language for comparing resolutions.

Refinement towers compare packet descriptions across levels. Weighted transfers and integral separation show which structures survive a change of resolution.

IN THIS CHAPTER

Mathematical text 99

Visual studies 152

Formal companion 1721

Sources and references 2668

Scope

Canonical values are defined for admissible data. A general existence theorem for admissible contours remains a separate question.

The next chapter moves from linear packet coordinates to typed nonlinear operations and analytic jets.

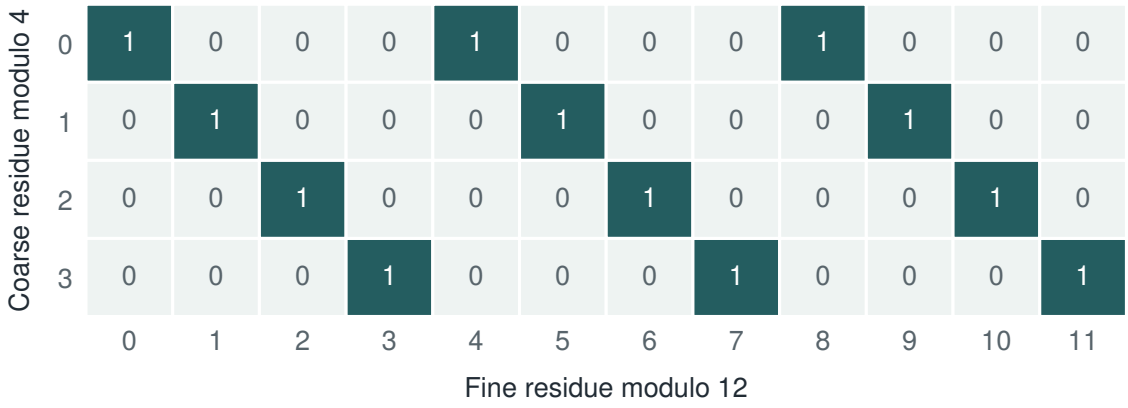


Figure 2. The incidence matrix $M_{r,s} = 1$ when the fine residue s belongs to the coarse residue r. Its rows implement exact aggregation from fine packets to coarse packets.

for all α and β. Remove the zero operators from the family

$$R_{\alpha\beta} = P_\alpha Q_\beta.$$

The remaining operators form a packet system that refines both $\mathcal{P}$ and $\mathcal{Q}$. The same statement holds for any finite collection of pairwise commuting packet systems.

Proof. Commutativity gives

$$R_{\alpha\beta}^2 = P_\alpha^2 Q_\beta^2 = R_{\alpha\beta}.$$

If $(\alpha, \beta) \neq (\alpha', \beta')$, then either $P_\alpha P_{\alpha'} = 0$ or $Q_\beta Q_{\beta'} = 0$, so the corresponding products annihilate one another. Finally,

$$\sum_{\alpha,\beta} R_{\alpha\beta} = \left(\sum_\alpha P_\alpha\right)\left(\sum_\beta Q_\beta\right) = I_X.$$

For a fixed α,

$$P_\alpha = P_\alpha \sum_\beta Q_\beta = \sum_\beta R_{\alpha\beta},$$

and the same argument works for a fixed β. Since each P_α and Q_β is nonzero, these identities ensure that every parent has a nonzero product child. The two parent maps on the supported product family are therefore surjective. Iteration proves the finite family statement. For the empty collection, use the supported identity family: (I_X) when $X \neq 0$ and the empty family when $X = 0$. □

Proposition 3.11 (Coarsest common refinement)**.** *Under the hypotheses of theorem 3.10, let $\mathcal{R}$ be a packet system that refines both $\mathcal{P}$ and $\mathcal{Q}$. Then $\mathcal{R}$ refines the nonzero product family $(P_\alpha Q_\beta)$. Hence that product family is the coarsest packet system that refines both $\mathcal{P}$ and $\mathcal{Q}$.*

CHAPTER III

Projector Resolved Seed Calculus III

Typed transfer, nonlinear packets, and obstruction theory

The preceding constructions provide the linear setting. This chapter studies nonlinear coordinates and the limits of their extension.

Typed operations extend the linear language to nonlinear coordinates, analytic packet jets and obstruction theory. The analytic supplement develops the continuous packet-jet claims.

IN THIS CHAPTER

Mathematical text 178

Analytic supplement 248

Visual studies 254

Formal companion 1768

Sources and references 2670

Scope

Expanded ordinary proofs and the bounded Lean core have different coverage. The packet Taylor estimates, counting arguments and later extensions retain their stated ordinary-proof status. Arbitrary differentiability is not asserted.

The next chapter develops a cyclic analytic family through coefficient selection, transport and spectral data.

Then

$$J_p = J_{a_n \leftarrow a_0}.$$

In particular, if $a_n = a_0$, *then* $J_p = \mathrm{id}$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.pathEndpoint;Komorebi.PaperIII.Transport.exactPathTransport;Komorebi.PaperIII.Transport.exactPathTransport_eq_endpoint;Komorebi.PaperIII.Transport.exactPathTransport_loop`

Proof. Induct on the list of successive vertices and apply the cocycle in theorem 4.1 at each step. If the endpoint is the start, the endpoint transport is the identity. □

4.2 Exact packet coordinates

The graph model above resolves a fixed scalar matrix. Coordinate transport is a separate construction. Let X be an R-module, and let

$$\mathcal{P}^a = (P_i^a)_{i \in I_a}$$

be a finite complete family of pairwise orthogonal linear idempotents:

$$P_i^a P_j^a = \delta_{ij} P_i^a, \qquad \sum_{i \in I_a} P_i^a = I_X.$$

Define the exact packet coordinate module

$$\mathrm{Coord}(\mathcal{P}^a) = \left\{ (x_i)_{i \in I_a} : P_i^a x_i = x_i \right\}.$$

Its analysis and synthesis maps are

$$\mathcal{A}_a(x) = (P_i^a x)_i, \qquad \mathcal{S}_a((x_i)_i) = \sum_i x_i.$$

Completeness and orthogonality give

$$\mathcal{S}_a \mathcal{A}_a = \mathrm{id}_X, \qquad \mathcal{A}_a \mathcal{S}_a = \mathrm{id}_{\mathrm{Coord}(\mathcal{P}^a)}.$$

What a packet coordinate contains

A packet coordinate is a whole vector fixed by its projector. Analysis splits an ambient vector into these components, and synthesis adds them back. The transport between exact systems keeps the ambient vector and changes its packet presentation.

Definition 4.3 (Canonical exact coordinate transport)**.** For two exact packet systems on the same ambient module, define

$$J^X_{b \leftarrow a} = \mathcal{A}_b \mathcal{S}_a : \mathrm{Coord}(\mathcal{P}^a) \longrightarrow \mathrm{Coord}(\mathcal{P}^b).$$

The arrow points from the source coordinate system a to the target coordinate system b.

CHAPTER IV

Projector Resolved Seed Calculus IV

Cyclic analytic functions, reciprocal families, and spectral zero data

The focus shifts from the general packet language to a particular analytic family.

A cyclic analytic family connects coefficient selection, differential transport, reciprocal polynomials and spectral zero data.

IN THIS CHAPTER

Mathematical text 280

Visual studies 355

Formal companion 1815

Sources and references 2672

Scope

The analytic statements retain their branch choices, multiplicities and integer-order hypotheses. Universal zero simplicity and a zeta functional equation remain outside these results.

The next chapter turns to signals, where filters, packet trees and reconstruction make changes of resolution concrete.

Fix any $\rho \in \mathbb{C}$ with $\rho^N = -1$.

Proposition 5.2 (Relation with the transfer exponential)**.** *For every j,*

$$\mathsf{D}_{N,j}(z) = \rho^{-j}\mathsf{E}_{N,j}(\rho z).$$

Proof. The coefficient of z^{Nn+j} on the right is

$$\rho^{-j}\rho^{Nn+j}\frac{1}{(Nn+j)!} = (-1)^n\frac{1}{(Nn+j)!}.$$

□

Theorem 5.3 (Signed derivative cycle)**.** *For $1 \le j < N$,*

$$\mathsf{D}'_{N,j} = \mathsf{D}_{N,j-1},$$

and

$$\mathsf{D}'_{N,0} = -\mathsf{D}_{N,N-1}.$$

Every signed section satisfies

$$\mathsf{D}^{(N)}_{N,j} = -\mathsf{D}_{N,j}.$$

The N signed sections form the basis of $y^{(N)} = -y$ determined by

$$\mathsf{D}^{(k)}_{N,j}(0) = \delta_{jk}$$

for $0 \le j, k < N$.

Proof. The derivative formulas follow directly from the series. The first nonzero term of $\mathsf{D}_{N,j}$ is $z^j/j!$, so the initial value matrix is the identity. The basis statement follows from the dimension of the solution space. □

Corollary 5.4 (Wronskian of the signed basis)**.** *Assume $N \ge 2$. The Wronskian of*

$$\mathsf{D}_{N,0}(z), \dots, \mathsf{D}_{N,N-1}(z)$$

is one.

Proof. The derivative matrix at the origin is the identity. The determinant factorization in the proof of corollary 4.3 applies with $a_\ell = \rho\omega^\ell$ and with the signed-section Fourier weights. Since $\sum_\ell a_\ell = 0$, the determinant is constant and equals one. □

Proposition 5.5 (Twisted addition law)**.** *For $0 \le k < N$,*

$$\mathsf{D}_{N,k}(z+w) = \sum_{\substack{0 \le i,j < N \\ i+j \equiv k \bmod N}} (-1)^{\lfloor (i+j)/N \rfloor}\mathsf{D}_{N,i}(z)\mathsf{D}_{N,j}(w).$$

Proof. Use proposition 5.2 and corollary 4.4. The factor introduced by a pair (i, j) is

$$\rho^{i+j-k}.$$

PLATE IV.4 / CYCLIC CONVOLUTION

Products mix the residue lanes

Chapter IV: Theorem 3.3.

Selection separates a function additively. It does not make multiplication diagonal. A residue-i term multiplied by a residue-j term lands in residue $i + j \pmod N$. For $N = 3$, use

$$F = 1 + 2z + 3z^2 + z^3, \qquad G = 2 + z + z^2.$$

The table displays every product of their three selected sections.

	2	z	z^2
$z^3 + 1$	$2z^3 + 2$	$z^4 + z$	$z^5 + z^2$
$2z$	$4z$	$2z^2$	$2z^3$
$3z^2$	$6z^2$	$3z^3$	$3z^4$

$$\mathcal{B}_{3,0}(FG) = 2 + 7z^3$$

shaded cells: $(0,0), (1,2), (2,1)$ all land in residue 0

The shaded cells contribute to residue zero:

$$2(1 + z^3) + (2z)(z^2) + (3z^2)z = 2 + 7z^3.$$

Keeping only the product of the residue-zero inputs would give $2 + 2z^3$ and lose five units of the z^3 coefficient. The mixed cells are part of the exact result.

The general rule

$$\mathcal{B}_{N,k}(FG) = \sum_{i+j \equiv k \ (N)} (\mathcal{B}_{N,i}F)(\mathcal{B}_{N,j}G).$$

Every pair belongs to exactly one output residue. No term is discarded and no coefficient is estimated. This finite cyclic convolution is the algebra behind the later addition formulas and block-coefficient recurrences.

> **Read this as:** A packet label tells you where an interaction lands, not that interactions with other packets vanish.

All nine table entries and all three output sections are checked as exact polynomial identities. This is the scalar cyclic specialization used by the paper.

CHAPTER V

Multiresolution Signal Calculus

Exact coordinates, singular packets, and stable reconstruction

The analytic constructions give way to signal representations and reconstruction.

Signal models move through filters, packet trees and reconstruction maps. Exact coordinates and singular packets distinguish representation from approximation and recovery.

IN THIS CHAPTER

Mathematical text 383

Visual studies 467

Formal companion 1936

Sources and references 2675

Scope

Separated-jump results require the stated model, gap and grid assumptions. Stable edge-packet recovery and the broader approximation class remain open.

The next chapter asks how transfers evolve in time and how eliminated states contribute memory.

Conversely, let $\boldsymbol{B} = \boldsymbol{R}\boldsymbol{E} - cz^{-d}I_N$ and suppose its action vanishes on every channel sequence. Insert an impulse at channel s, address zero, with every other channel zero. Output channel k reads the coefficients of the Laurent polynomial B_{ks} at all addresses. All of these coefficients vanish, so $B_{ks} = 0$ for every k, s. The representation is faithful, proving necessity as a polynomial identity rather than merely equality at one evaluation point. The displayed unit-gain case is $c = 1$. □

Original-sample delay. For an arbitrary original-sample delay $D \in \mathbb{Z}$, the output phase k has

$$y_k[n] = x_{r_k}[n + q_k], \qquad r_k = (k - D) \bmod N, \qquad q_k = (k - D) \operatorname{div} N.$$

Its channel matrix has the sole nonzero entry z^{q_k} in row k, column r_k. In particular, a one-sample delay at $N = 2$ has matrix

$$\begin{pmatrix} 0 & z^{-1} \\ 1 & 0 \end{pmatrix},$$

whereas $z^{-1}I_2$ delays the original signal by two samples. The general address identity is valid for negative sample addresses as well.

4.1 The Haar bank

The Haar analysis matrix is

$$\boldsymbol{E}_{\mathrm{H}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It is orthogonal, so $\boldsymbol{R}_{\mathrm{H}} = \boldsymbol{E}_{\mathrm{H}}^{\mathsf{T}}$ and

$$\boldsymbol{R}_{\mathrm{H}}\boldsymbol{E}_{\mathrm{H}} = I_2.$$

The channels are

$$a[n] = \frac{x[2n] + x[2n+1]}{\sqrt{2}}, \qquad d[n] = \frac{x[2n] - x[2n+1]}{\sqrt{2}}.$$

Corollary 4.2 (Haar energy identity)**.**

$$\boxed{\|x\|_2^2 = \|a\|_2^2 + \|d\|_2^2.}$$

Proof. For any $u, v \in \mathbb{C}$, the parallelogram identity gives

$$\left|\frac{u+v}{\sqrt{2}}\right|^2 + \left|\frac{u-v}{\sqrt{2}}\right|^2 = |u|^2 + |v|^2.$$

Apply this to $u = x[2n]$, $v = x[2n+1]$ and sum over $n \in \mathbb{Z}$. Each summand is nonnegative. The right side is $\|x\|_2^2$ by the two-channel address equivalence. This also proves that the literal low and detail channels belong to ℓ^2. The real Haar matrix is unitary over $\mathbb{C}$, so the finite-signal statement is the same calculation with a finite sum. □

4.2 Predict–update lifting

Split x into even and odd channels e, o. Define

$$d = o - e, \qquad s = e + \frac{1}{2}d.$$

CHAPTER VI

Transfer Semigroups and Memory

Probabilistic refinement, fractional operators, and exact hidden-state elimination

Static descriptions now become evolution and conditioning problems.

Evolution, probability and conditioning share a transfer language. Eliminating hidden states produces memory terms whose hypotheses remain part of the model.

IN THIS CHAPTER

Mathematical text 511

Visual studies 616

Formal companion 1992

Sources and references 2678

Scope

Finite-state, common-gap and fixed-partition restrictions remain explicit. Completion, uniform refinement control and stable infinite latent-packet closure remain open. Floating residuals are not interval certificates.

The next chapter treats quantum seeds, operator grammars and finite effective dynamics.

Theorem 3.3 (Resolvent refinement identity)**.**

$$\boxed{CR_f(z) - R_c(z)C = R_c(z)\Delta R_f(z).}$$

Proof. Multiply the left side by $zI - A_c$ on the left and $zI - A_f$ on the right. The result is

$$(zI - A_c)C - C(zI - A_f) = CA_f - A_cC = \Delta.$$

Multiplication by the two resolvents recovers the stated identity. □

4 Markov seeds and stochastic transfer

Use column probability vectors. A finite continuous-time Markov generator Q has nonnegative off-diagonal entries and column sums zero. The stochastic semigroup is

$$K(t) = e^{tQ}.$$

Theorem 4.1 (Finite Markov semigroup)**.** *If Q has nonnegative off-diagonal entries and column sums zero, then e^{tQ} has nonnegative entries and column sums one for every $t \geq 0$.*

Proof. Choose $\lambda \geq \max_i(-Q_{ii})$. Then $B = Q + \lambda I$ has nonnegative entries, so e^{tB} is entrywise nonnegative by its power series. Hence

$$e^{tQ} = e^{-\lambda t}e^{tB}$$

is nonnegative. Since $1^TQ = 0$, differentiation gives

$$\frac{d}{dt}1^Te^{tQ} = 1^TQe^{tQ} = 0,$$

so $1^Te^{tQ} = 1^T$. □

Example 4.2 (Symmetric random walk)**.** Starting from δ_0 and applying the transfer

$$T = \frac{1}{2}(S + S^{-1})$$

for n steps gives

$$\mathbb{P}(X_n = k) = 2^{-n}\binom{n}{(n+k)/2}$$

when $k \equiv n \pmod 2$, and zero otherwise. Hence

$$\mathbb{E}[X_n] = 0, \qquad \mathrm{Var}(X_n) = n.$$

The parity packet is exact: all mass remains in the one residue class permitted by n.

CHAPTER VII

Quantum Seeds and Effective Dynamics

Symmetry, geometric transport, quantum channels, and bounded refinement towers

The transfer viewpoint is now developed in quantum settings with their own algebraic and dynamical assumptions.

Quantum seeds generate spaces through an operator grammar. Symmetry, circuits, geometric transport and quantum channels lead to finite effective dynamics and bounded refinement questions.

IN THIS CHAPTER

Mathematical text 660

Visual studies 772

Formal companion 2054

Sources and references 2681

Scope

The finite formal contract does not certify the full continuum physics program. Anomaly-free refinement, uniform Feshbach control, Lorentzian reconstruction and the proposed gravitational limits remain open.

The next chapter turns to algorithms, separating exact structural reductions from numerical and hardware questions.

I.3.2 Tensor refinement

Fix $N \geq 1$. Suppose every factor a has a complete orthogonal family $P_0^{(a)}, \dots, P_{N-1}^{(a)}$. The fine tensor packets are

$$P_{\mathbf{j}}^{\text{fine}} = P_{j_1}^{(1)} \otimes \cdots \otimes P_{j_m}^{(m)}.$$

Theorem I.3.2 (Tensor/polyphase refinement). *For each residue $r \in \mathbb{Z}/N\mathbb{Z}$, define*

$$\boxed{P_r^{\text{coarse}} = \sum_{j_1+\cdots+j_m \equiv r \pmod N} P_{j_1}^{(1)} \otimes \cdots \otimes P_{j_m}^{(m)}.}$$

Then the P_r^{coarse} are complete orthogonal projectors, and every coarse packet is the exact sum of its fine descendants.

Proof. The fine tensor projectors are mutually orthogonal and sum to the identity. Grouping the fine addresses by their total residue partitions the full address set. Sums over disjoint sets of orthogonal projectors remain orthogonal projectors, and the union of the residue classes is the full fine address set. □

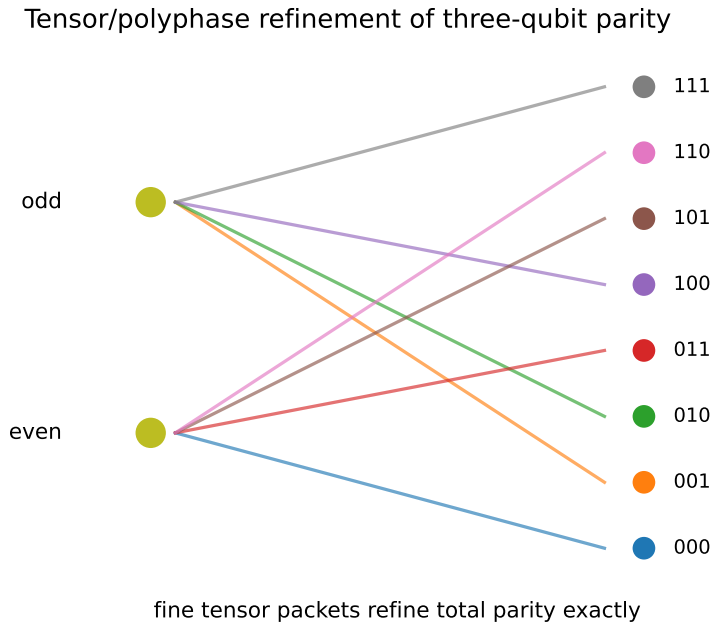


Figure 5. The eight computational packets of three qubits refine the two total-parity packets exactly.

Recorded Numerical Check I.3.3 (Exact parity refinement). For three qubits, the local computational projectors were grouped by the parity of their Hamming weight. The resulting coarse projectors agreed with $(I \pm Z^{\otimes 3})/2$ with residual 0.

CHAPTER VIII

Packet Operator Algorithms

Exact compilation and GPU realization

The earlier packet constructions become inputs to exact compilation and numerical implementation.

Structured packet representations remove work before execution. Exact compilation connects algebraic structure, recompression, precision and stated hardware observations.

IN THIS CHAPTER

Mathematical text 814

Visual studies 897

Formal companion 2132

Sources and references 2684

Scope

Small-core optimality uses supplied ordered thin-SVD equations and actual compact ranges. General numerical SVD verification, unrestricted discovery and CUDA-kernel verification require further results.

The book then turns to arithmetic, beginning with Euler congruences, completion and dyadic geometry.

2 Packet operator representations

2.1 Exact semantics

Definition 2.1 (Packet layout)**.** A packet layout is a finite family of finite dimensional Hilbert spaces

$$\mathcal{P} = (X_\alpha)_{\alpha \in I}$$

with direct sum

$$X = \bigoplus_{\alpha \in I} X_\alpha.$$

Write

$$n_\alpha = \dim X_\alpha, \qquad n = \sum_{\alpha \in I} n_\alpha.$$

The coordinate inclusion and projection are $\iota_\alpha : X_\alpha \to X$ and $\pi_\alpha : X \to X_\alpha$. The ambient coordinate projector is $P_\alpha = \iota_\alpha \pi_\alpha$.

Definition 2.2 (Packet operator representation)**.** A finite packet operator representation consists of

$$\mathsf{IR} = (\mathcal{P}, E, (B_{\beta\alpha})_{(\beta,\alpha)\in E}),$$

where

$$E \subseteq I \times I$$

and

$$B_{\beta\alpha} : X_\alpha \longrightarrow X_\beta.$$

Local blocks are identified with their ambient extensions by inclusion and projection. In particular, the notation $P_\beta A P_\alpha$ denotes either the ambient block or its restriction $X_\alpha \to X_\beta$ according to the displayed domain, and $\iota_\beta B_{\beta\alpha} \pi_\alpha$ is the corresponding ambient extension. Its evaluation is

$$\operatorname{eval}(\mathsf{IR}) = \sum_{(\beta,\alpha)\in E} \iota_\beta B_{\beta\alpha} \pi_\alpha,$$

where $\iota_\beta : X_\beta \to X$ is the coordinate inclusion.

Theorem 2.3 (Exact evaluation)**.** *Let $A \in \operatorname{End}(X)$. Set*

$$B_{\beta\alpha} = \pi_\beta A \iota_\alpha$$

and let E contain exactly the nonzero blocks. Then

$$\boxed{A = \sum_{(\beta,\alpha)\in E} P_\beta A P_\alpha.}$$

How to use this volume

Follow each small operator through its factors and formulas, then read the cited general argument in the traditional volume. The Lean companion records the separate finite formal contract.

Routes through the guide

Plates VIII.1 through VIII.32 identify worked pages independently of PDF pagination. Each source line names the corresponding general statement in the mathematical text. The integrated atlas uses the same plate labels.

Route	Mathematical question	Plates
Representations	Which exact data represent the required operator?	1–4, 27
Structured transforms	Which conventions preserve the same map?	5–8
Low-rank algebra	What remains exact after products and recompression?	9–13, 29
Support and solves	Which interactions can occur or cancel?	14–18, 30
Precision and compilation	Which finite costs and errors does the optimizer use?	19–23, 31–32
Performance contracts	Which one-time costs and outputs are compared?	24–26
Evidence	What is checked, proved, or still open?	2, 28

Keep the represented object visible

A dense array, factored operator, and execution plan may describe the same map while returning different objects. Representation savings are exact modeled counts. Measured speed requires a separate experiment with matching output and timing contracts.

Preservation and evidence

These plates illustrate the chapter's finite constructions. The accompanying models and checking code distinguish exact examples, symbolic calculations, bounded enumerations and rejected controls. Drawing geometry and model checks have separate scopes.

The finite formal correspondence contains 56 manuscript-proved items, 392 named declarations in 40 modules and 66 construction audits. Nineteen declarations are reused foundation results. The ordered-SVD hypotheses, four classical entries, recorded timing corpus and three broad research targets retain their separate scopes. These records do not claim a new Lean compiler execution.

Read this as: Choose a question, calculate its small model, and check the hypotheses, output contract, and evidence level of the corresponding general result.

CHAPTER IX

Euler Congruences

Completion, factorial strengthening, and sharp dyadic geometry

The arithmetic sequence begins with congruence identities and the hypotheses needed to extend them.

Euler congruences are developed through completion, factorial strengthening and dyadic geometry. The accompanying guides connect exact statements, bounded examples and formal coverage.

IN THIS CHAPTER

Mathematical text 931

Visual studies 1145

Formal companion 2220

Sources and references 2687

Scope

The arithmetic, dyadic and lifting results have distinct hypotheses. General Euler-tail simplicity and the broader native and polyphase extensions remain open.

The next chapter organizes residual seeds and conditional system completion from supplied arithmetic data.

The two factors are uniquely determined by these properties and are compatible with reduction to lower precision. For $a \in (\mathbb{Z}/2^{L+2}\mathbb{Z})^\times$, there is instead a unique decomposition

$$a = \epsilon(a)\langle a\rangle_4, \qquad \epsilon(a) \in \{1,-1\}, \qquad \langle a\rangle_4 \equiv 1 \pmod 4.$$

Proof. The unit-group exponent $p^L(p-1)$ gives the root identity for ω_L. Its residue is that of a, so the remaining factor has residue 1. If two $(p-1)$st roots have equal residue, raising them to p^L makes them equal modulo p^{L+1}, while each root is fixed by this exponent. This proves uniqueness and then compatibility under reduction. In the dyadic construction, an odd residue is exactly 1 or -1 modulo 4. Choose that sign and multiply a by it to obtain the principal factor. The two signs remain distinct modulo 4, which proves uniqueness. □

Formal correspondence. `FiniteUnits.finite_unit_decomposition`, `FiniteUnits.decomposition_unique`, `UnitRefinement.omega_compatible`, `UnitRefinement.principal_compatible`, and `DyadicUnits.dyadic_unit_decomposition` with `DyadicUnits.decomposition_unique`. The finite $p = 2$ version of the first construction has trivial root-of-unity factor. The separate modulo-4 construction retains the sign that reduction modulo 2 cannot detect.

2.6 Recurrence packets and finite arithmetic synchronization

Theorem 2.10 (Companion jumps and packet recurrences)**.** *Let C be a $d \times d$ matrix over a commutative ring and $x_n = C^n x_0$. For $N, j, k \in \mathbb{N}$, put $b_k = x_{Nk+j}$. Then*

$$b_k = C^j (C^N)^k x_0, \qquad b_{k+1} = C^N b_k.$$

If $\chi_{C^N}(X) = X^d + \sum_{r<d} q_r X^r$, then

$$b_{k+d} = -\sum_{r=0}^{d-1} q_r b_{k+r}.$$

Every coordinate and every linear scalar observable satisfies the same recurrence. No invertibility or diagonalization hypothesis is required.

Proof. The jump identity follows from the power law for one matrix. Apply the Cayley–Hamilton identity $\chi_{C^N}(C^N) = 0$ to the state b_k. The leading coefficient is 1, so the final state can be isolated. A linear observable preserves the resulting finite sum. □

Formal correspondence. `Companion.companion_jump`, `Companion.packet_charpoly_recurrence`, and `Companion.packet_observable_recurrence`. The source-sequence versions derive the packet recurrence from the relation $x_{n+1} = Cx_n$. For the Fibonacci matrix $C = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, its jump trace is the Lucas number L_N and its determinant is $(-1)^N$. Thus

$$F_{N(k+2)+j} = L_N F_{N(k+1)+j} - (-1)^N F_{Nk+j}.$$

This specialization is `FibonacciPackets.fibonacci_packet_recurrence`. It is derived from the Fibonacci state.

Theorem 2.11 (Dyadic–odd packet factorization)**.** *Let $a \geq 0$ and let M be a positive odd integer. The Chinese remainder map is a ring isomorphism*

$$\mathbb{Z}/(2^a M)\mathbb{Z} \simeq \mathbb{Z}/2^a\mathbb{Z} \times \mathbb{Z}/M\mathbb{Z}.$$

For $0 \leq r < 2^a M$ and any coefficient sequence f,

$$P_{2^a M, r} f = P_{2^a, r \bmod 2^a}\, P_{M, r \bmod M} f.$$

CHAPTER X

Universal Residual Seeds

Deformation forests, transport categories, and conditional system completion

The congruence study is followed by a framework for residual seeds and compatible arithmetic descriptions.

Residual seeds, deformation forests and transport categories organize conditional system completion. Supplied arithmetic and carrier data remain explicit inputs.

IN THIS CHAPTER

Mathematical text 1170

Visual studies 1227

Formal companion 2350

Sources and references 2690

Scope

The target family and recovered carrier are independent existential outputs. Identifying them and proving nonzero occupancy require additional premises. Source families, carrier presentations and local comparisons remain supplied.

The next chapter studies local packet geometry and the remaining obligations of arithmetic realization.

4 Tangent and obstruction packets

For a residual representation $\bar{\rho}$, first-order deformations and lifting obstructions are classically controlled by cohomology of the adjoint module. The projector calculus does not replace that theory. It resolves the controlling complex into invariant packets when suitable projectors exist.

Classical Input 4.1 (Tangent and obstruction spaces)**.** For a framed representation problem, first-order lifts are described by $Z^1(G, \operatorname{ad}\bar{\rho})$. For an unframed Schur problem, tangent directions are described by $H^1(G, \operatorname{ad}\bar{\rho})$. Relations in a presentation of a deformation ring map into $H^2(G, \operatorname{ad}\bar{\rho})$, which contains the obstruction classes.

Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2 \to \cdots$$

be a cochain complex over a field. Suppose a finite family of projectors $P_j^r : C^r \to C^r$ satisfies

$$\sum_j P_j^r = I, \qquad P_i^r P_j^r = \delta_{ij} P_j^r, \qquad d_r P_j^r = P_j^{r+1} d_r.$$

Theorem 4.2 (Cohomology packet decomposition)**.** *Under these hypotheses, each packet is a subcomplex and*

$$\boxed{H^r(C^\bullet) \cong \bigoplus_j H^r(P_j C^\bullet).}$$

In particular,

$$H^1 = \bigoplus_j H^1_j, \qquad H^2 = \bigoplus_j H^2_j,$$

so tangent and obstruction spaces reconstruct exactly from their packet components.

Proof. For each packet, use the actual image submodules $C_j^r = \operatorname{im} P_j^r$. Commutation with the differential makes these a subcomplex. Write

$$Z^r = \ker d_r, \qquad B^r = \operatorname{im} d_{r-1} \subseteq Z^r, \qquad H^r = Z^r / B^r,$$

and define Z_j^r, B_j^r, H_j^r using the restricted differentials of $C_j^\bullet$. The inclusion $i_j : C_j^\bullet \to C^\bullet$ and the image-valued projection $p_j : C^\bullet \to C_j^\bullet$ are cochain maps, so they induce maps on these genuine quotients.

For $i \neq j$, orthogonality gives $H(p_i)H(i_j) = 0$. Idempotence gives $H(p_j)H(i_j) = I$, and completeness gives $\sum_j H(i_j)H(p_j) = I$. Thus the maps

$$[z] \longmapsto ([P_j^r z])_j, \qquad ([z_j])_j \longmapsto \left[\sum_j i_j z_j\right]$$

are inverse linear maps. The finite product equals the direct sum, proving the assertion. In the formalization, both sides are also identified with Mathlib's standard homology objects of the genuine graded complexes. □

Scope of the splitting. The algebraic proof works over any commutative ring and does not divide by 2, so it applies in characteristic two when the stated integral chain projectors exist. It does not construct

CHAPTER XI

Tame and Wild Packet Geometry

For local Langlands

The preceding chapter separates the arithmetic objects supplied as inputs from those obtained conditionally. Local packet geometry sharpens the realization questions.

Tame and wild packet geometry organizes the local arithmetic constructions. The research companions retain both restricted progress and the unresolved obligations of Conjecture 66.

IN THIS CHAPTER

Mathematical text 1260

Visual studies 1367

Formal companion 2384

Further arithmetic studies 1482

Sources and references 2693

Scope

Conjecture 66 remains open. Its unresolved requirements concern general arithmetic realization, H5 for the same candidate, corrected full-character occurrence and actual native data under the stronger contract.

The final chapter gives a conditional route through patching support and Hecke compatibility. The open realization obligations remain separate.

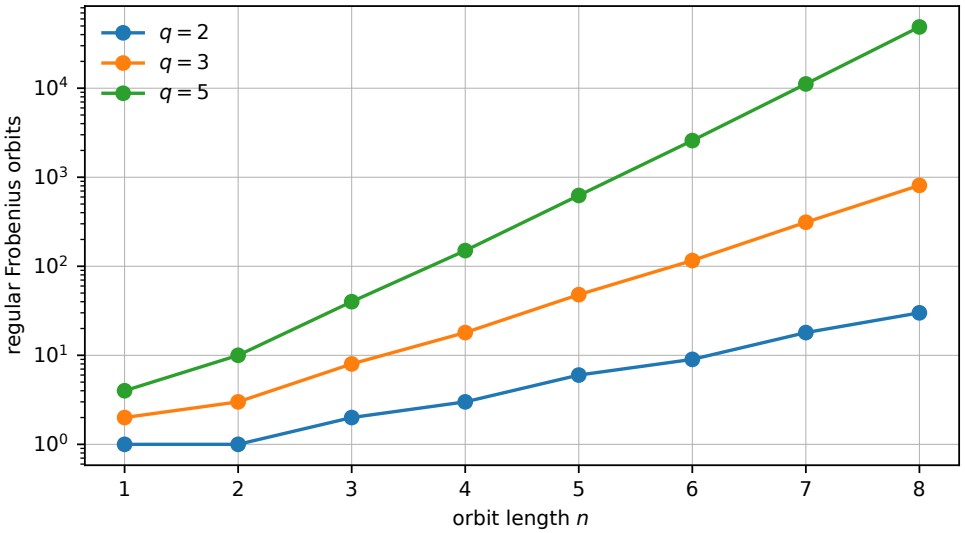


Figure 4. Regular Frobenius orbit counts for three residue field sizes.

The formula gives

$$\frac{1}{4}\big((2^4-1)-(2^2-1)\big)=3.$$

4 Finite field cuspidal projectors

Let

$$G_n=\mathrm{GL}_n(\mathbb{F}_q),\qquad T_n\cong\mathbb{F}_{q^n}^{\times}$$

with T_n an elliptic maximal torus.

Standard Fact 4.1 (Regular elliptic torus characters)**.** A regular character $\theta : T_n \to \mathbb{C}^{\times}$ determines an irreducible cuspidal representation κ_θ of G_n. Frobenius conjugate characters determine isomorphic representations, and every irreducible cuspidal representation of G_n occurs in this way.

This is the finite field input from Deligne and Lusztig theory [13]. We use standard finite group character theory for the central idempotents [28, 21]. Once κ_θ is known, its block is selected by a central idempotent.

Proposition 4.2 (Primitive cuspidal idempotent)**.** *Let χ_θ be the irreducible character of κ_θ, and let $d_\theta=\chi_\theta(1)$. Then*

$$e_\theta=\frac{d_\theta}{|G_n|}\sum_{g\in G_n}\chi_\theta(g^{-1})g$$

is the primitive central idempotent selecting the κ_θ block in $\mathbb{C}[G_n]$.

Proof. On an irreducible representation of character ψ, the central element acts by a scalar. Taking traces shows that the scalar is

$$\frac{d_\theta}{|G_n|\,\psi(1)}\sum_{g\in G_n}\chi_\theta(g^{-1})\psi(g).$$

Character orthogonality makes this scalar one for $\psi=\chi_\theta$ and zero otherwise. The element is therefore central, idempotent, and primitive. □

CHAPTER XII

Patching Support and Hecke Packets

Conditional automorphic completion

The arithmetic sequence culminates in a conditional argument using patching support, Hecke compatibility and the stated representation-theoretic inputs.

Patching support and compatible Hecke packets complete a conditional arithmetic argument. Each application must supply the geometric, automorphic and representation-theoretic inputs that the theorem uses.

IN THIS CHAPTER

Mathematical text 1418

Visual studies 1455

Formal companion 2617

Sources and references 2702

Scope

Actual residual and deformation data, applicable patching and full support, a compatible same-field Hecke point, classicality and an attached representation are required. Comparisons of compatible families assume those families already exist. Chapter XI's Conjecture 66 is not assumed proved.

The formal companions at the back of the book describe their own precise coverage. The glossary provides a route back to the principal definitions.

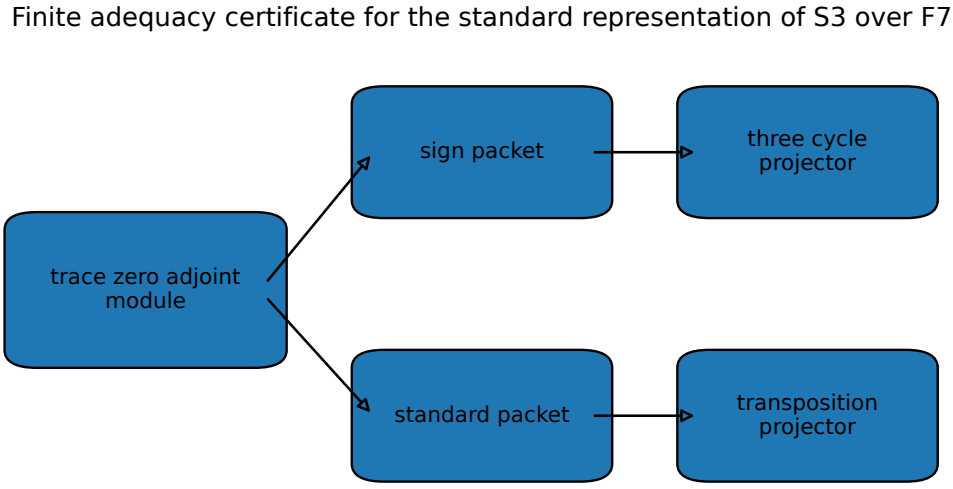


Figure 4. The two irreducible adjoint packets are detected by different generalized eigenspace projectors.

Its image equals $\ker((D-\alpha I)^n)$ for every $n \geq 1$, and it kills $\ker((D-\beta I)^n)$. This identifies the displayed matrices with the generalized eigenspace projectors required in the adequacy definition.

For detection on every irreducible submodule, one can check the stronger assertion that canonical projectors detect every nonzero matrix. In addition to the two projectors above, use

$$P_{s,-1} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}, \qquad P_{tst^{-1},1} = \begin{pmatrix} 1 & 0 \\ 4 & 0 \end{pmatrix}.$$

The four trace pairings with these matrices have trivial common kernel in $M_2(\mathbb{F}_7)$, by row reduction. Every irreducible adjoint submodule contains a nonzero matrix, so some canonical projector has nonzero trace pairing on that submodule. This verifies the last adequacy clause and completes the certificate. □

Remark 4.3. The theorem verifies adequacy for one finite image. A global automorphy lifting argument must verify the published residual image hypothesis for its actual representation. The calculation does not replace that step, and it does not provide a uniform coefficient prime two theorem.

5 Auxiliary group algebras and augmentation

Let Δ be a finite abelian p group and let

$$\Lambda = \mathcal{O}[\Delta].$$

The augmentation map is

$$\epsilon : \Lambda \longrightarrow \mathcal{O}, \qquad \sum_{g \in \Delta} a_g g \longmapsto \sum_{g \in \Delta} a_g.$$

Its kernel is the augmentation ideal $\mathfrak{a}$.

Proposition 5.1 (Functorial group algebra transfer)**.** *A group homomorphism $q : \Delta' \to \Delta$ induces an $\mathcal{O}$ algebra homomorphism*

$$q_* : \mathcal{O}[\Delta'] \longrightarrow \mathcal{O}[\Delta].$$

Further studies

The following studies develop the restricted arithmetic constructions, obstructions and remaining realization questions associated with Chapter XI.

READING ROUTES

Arithmetic realization and its obstructions 1483

Visual studies of arithmetic realization 1551

FURTHER STUDY

Arithmetic realization and its obstructions

These studies distinguish finite packet constructions, actual arithmetic candidates, full-character occurrence and comparison for the same candidate.

READING ROUTES

Research companion 1484

Scope

Conjecture 66 remains open. Its unresolved requirements concern general arithmetic realization, H5 for the same candidate, corrected full-character occurrence and actual native data under the stronger contract.

6.2 Information tests do not replace full pair factors

The inherited rank-nine information test [10] is more specific. Its rank-nine examples, conditional on the pinned source construction used there, have the same recorded determinant, Frobenius matrix, slope, Swan conductor, projective Weil map and self-pair data. They differ in an actual scalar quotient map and are separated by a stated twist test. In the recorded test the Artin conductors are 33 and 45. The source does not claim equality of H5, and these examples are not counterexamples to local-factor characterization [10].

Twist transport has a positive use as well. Once a compatible base realization is genuinely supplied, its character and inducing data can be transported under the stated twist. That transports an existing witness. It does not supply the missing base witness. The distinction is the same one encountered in the formal existence field.

6.3 Equal numerical profiles do not determine quotient capacity

The later dyadic record gives an exact arithmetic comparison of two host patterns. At rank eight with $(h_1, h_2, h_3) = (1, 3, 5)$, the full defining tower has levels $(60, 18, 3)$, Swan conductor 60, torsion number 1 and self-pair conductor 342. Its symplectic type quotient has dimension 6. A skipped-degree pattern $8 \to 2$ can have the same three numerical invariants with quotient dimension 4 [18].

The formula makes the comparison transparent. Put $L = 12h_3$ and $c = h_1 + 4h_2$. For the stated skipped-degree construction,

$$a_{\mathrm{skip}} = 63 + 4L + 3c = 63 + 48h_3 + 12h_2 + 3h_1 = a_{\mathrm{full}}.$$

Yet the two capacity formulas remain different. The recorded exact checks verify the displayed arithmetic identities. The assertion that both patterns arise from actual simple strata remains an ordinary arithmetic claim established in the retained source.

7 The direct odd-prime construction

Evidence and scope. The odd-prime construction [11–14] uses ordinary parameter and character arguments relative to the explicit Bushnell–Kutzko inputs in Section 13. The cited statements and their principal hypotheses were checked in the recorded source audit. This supplies neither Lean proofs of simple-character theory nor a proof of H5.

7.1 Notation and the construction contract

Let p be odd and put $F = \mathbb{F}_p((t))$. Choose

$$0 < h_1 < \cdots < h_d, \qquad p \nmid h_i, \qquad q_i = (p+1)h_i, \qquad N = p^d.$$

The construction produces a literal matrix $\beta \in M_N(F)$, rather than a formal symbol for an unknown arithmetic parameter. Its generated field is maximal and totally ramified, and it normalizes a complete lattice chain. A subsequent, separate argument constructs a full simple character and an extended inducing representation. Only after those steps is there an actual supercuspidal candidate.

At an intermediate local field L, write $\mathcal{O}_L$ for its ring of integers and t_L for its chosen uniformizer. For a hereditary order $\mathfrak{A}$ with radical $\mathfrak{P}$, put $U^r_{\mathfrak{A}} = 1 + \mathfrak{P}^r$ for $r \geq 1$. The valuation $v_{\mathfrak{A}}(x)$ is the largest integer r with $x \in \mathfrak{P}^r$. On a complete period-N chain, scalar valuations satisfy $v_{\mathfrak{A}}(a) = N v_L(a)$. We use ordinary matrix trace over the displayed coefficient field. No trace is divided by a matrix size.

FURTHER STUDY

Visual studies of arithmetic realization

The plates give worked constructions and counterexamples, with explicit links to the corresponding arguments and their hypotheses.

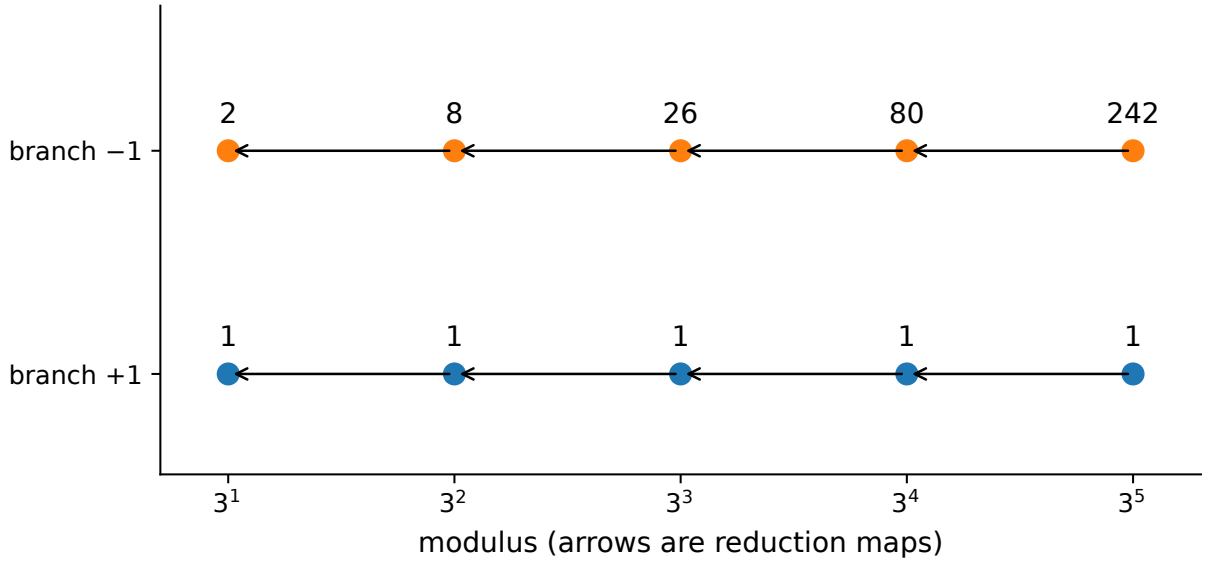
READING ROUTES

Research atlas 1552

PLATE XI.4

What a compact binder can do

Compatible solutions in one fixed compact space can survive every precision.



The points are all roots of $x^2 - 1$ modulo 3^n, for $1 \le n \le 5$. The labels are canonical residue representatives. The arrows reduce a root modulo 3^n to a root modulo 3^{n-1}. Both branches persist, corresponding to 1 and -1 in $\mathbf{Z}_3$.

For the general binder, fix a closed $X \subseteq \mathcal{O}^m$ and continuous constraints f_j. The sets

$$C_n = \{x \in X : f_j(x) \in \varpi^n \mathcal{O} \text{ for every } j \le n\}$$

are closed and nested. If every C_n is nonempty, compactness supplies a point satisfying every exact equation.

The example does not prove simultaneous finite solvability for the arithmetic problem. Even $x = 0$ and $x = 1$ have separately solvable congruences but no joint solution modulo three. Allowing the number of coefficients or pole bounds to grow also changes the compact model. Those missing hypotheses cannot be replaced by more isolated successful tests.

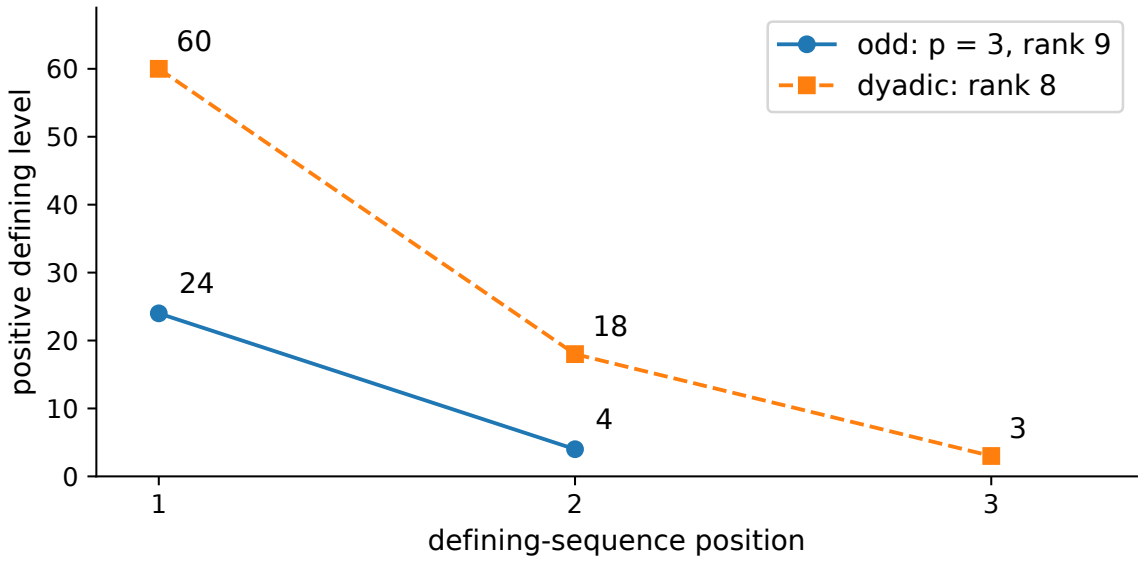
Evidence: Exact finite illustration.

Companion source: Part I, Section 5; Part II, Section 19.1. Book page 1490.

PLATE XI.8

Two defining-sequence examples

The recursion records levels and field refinements, not just its final matrix size.



The odd example uses $p = 3$, $d = 2$, and $(h_1, h_2) = (1, 2)$. Its displayed levels are $(24, 4)$. The dyadic example uses $d = 3$ and $(h_1, h_2, h_3) = (1, 3, 5)$, giving $(60, 18, 3)$. Each position refers to that example's own defining sequence.

$$\nu_j = p^{d-j}(p+1)h_{d-j+1} \quad \text{or} \quad \nu_j = 2^{d-j}3h_{d-j+1}.$$

The two plotted lines are not empirical fits or samples of one common arithmetic family. They make the inherited level formulas legible. In the odd example, $24 > 3 \cdot 4 > 2 \cdot 4$ illustrates the separated scales used in refinement.

Supplying the levels does not prove that the literal corrected matrix generates the asserted maximal field. That conclusion uses the filtered corestriction lift and the precise refinement theorem. In the dyadic argument, separability from native trace one is established only after the full field degree has been obtained.

Evidence: Inherited formulas, exact evaluation.

Companion source: Part I, Sections 7.1–7.4 and 8.1–8.3. Book page 1492.

Formal companions

The formal companions connect the mathematical chapters to their Lean declarations, hypotheses, source correspondences and verification evidence. Read each statement with the coverage described by its companion.

READING ROUTES

I Projector Resolved Seed Calculus I 1588

II Projector Resolved Seed Calculus II 1721

III Projector Resolved Seed Calculus III 1768

IV Projector Resolved Seed Calculus IV 1815

V Multiresolution Signal Calculus 1936

VI Transfer Semigroups and Memory 1992

VII Quantum Seeds and Effective Dynamics 2054

VIII Packet Operator Algorithms 2132

IX Euler Congruences 2220

X Universal Residual Seeds 2350

XI Tame and Wild Packet Geometry 2384

XII Patching Support and Hecke Packets 2617

Projector Resolved Seed Calculus I

The formal companion to Chapter I explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Computational formulation 1589

Formal companion 1683

Return to Chapter I

Their sum is T. The exact reconstruction API returns:

```
{
  "operator": [
    [
      "2",
      "3"
    ],
    [
      "5",
      "7"
    ]
  ]
}
```

The input and output JSON files for this reconstruction are supplied in `examples/`. The reconstruction action performs the block sum. Its interpretation as recovery of the original operator uses the validated packet system and the way those blocks were formed.

5.3 Composing resolved operators

Now set

$$S = \begin{pmatrix} 11 & 13 \\ 17 & 19 \end{pmatrix}, \qquad ST = \begin{pmatrix} 87 & 124 \\ 129 & 184 \end{pmatrix}.$$

The runtime forms the blocks of S and T, then sums over the middle packet index. It compares the resulting array with the directly resolved product. The supplied worked output has `equals_direct: true`. The full block arrays are retained in `examples/composition.json`.

The comparison checks a concrete instance of the multiplication identity. The general identity follows from inserting the complete family between the factors, as in the source proof reproduced in Appendix A.

5.4 An intertwining computation

Let

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad Q_i = UP_iU^{-1}, \qquad S' = UTU^{-1} = \begin{pmatrix} 7 & 3 \\ 5 & 2 \end{pmatrix}.$$

The target packets are $Q_0 = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 0 \end{smallmatrix}\right)$ and $Q_1 = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. The global identity is $UT = S'U$.

The recorded execution of `LOWER_INTERTWINER` returns true for both the global identity and the conjunction of the four local equations. Each local residual is the zero matrix. One complete local result is:

```
{
  "holds": true,
  "lhs": [
    [
      "-3",
      "3"
    ],
    [
      "0",
      "0"
    ]
  ],
  "residual": [
    [
```

2 Formal sources, correspondence, and evidence

The mathematical text and both companions share one statement inventory. The traditional manuscript supplies the statements and proofs. The machine catalog reproduces those statements with stable identifiers and source locators. The formal correspondence map records the exact Lean support for each statement. The source archive uses the following stable paths:

File or directory	Purpose
`formal/current/`	Flat formal source closure, including the continuation modules and their earlier dependencies.
`formal/coverage/COVERAGE_108.json`	All 108 statements, source proofs and hashes, declaration mappings, scopes, and record classifications.
`formal/evidence/`	Compiler, replay, audit, identity, and boundary-check receipts, indexed by the verification record.
`editable/traditional/`	Mathematical source, including the stated hypotheses of the split-semisimple corollary.

The inventory has 69 proof-level records. Definitions, classical inputs, boundaries, and documentary records are listed separately in the map and are excluded from this proof-level count.

Status	Covered	Partial	Not mapped
Proof-level records	69	0	0

"Covered" means a complete correspondence at the recorded scope. "Partial" includes substantive missing proofs and consequences not yet assembled from existing declarations. "Not mapped" means that this source closure has no established complete declaration for the record. These statuses are not a percentage of independent theorems: the records differ in scope and may share mathematical dependencies.

2.1 Primary and semisimple constructions

The continuation proves the intrinsic primary quotient and generator count, the geometric formula, scalar extension, and the six-way cyclic criterion. It also proves noncommutative module-top reduction and the nonsplit semisimple formula, including actual compatible decompositions from the Wedderburn–Artin existence theorem. The final existence theorems do not take a desired rank equality or a compatible decomposition as an input. They are noncomputable mathematical constructions, not executable decomposition algorithms.

Packet Operator Algorithms

The formal companion to Chapter VIII explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2133

Return to Chapter VIII

4 Representation algebra and declared cost models

4.1 Why a count theorem can have a short proof

`PacketCostModels` defines finite address and event types before counting them. The block widths are natural numbers. A diagonal block has a pair of within-block indices. A thin factor has one ambient and one rank index. An application event additionally has a right-hand-side index and a Boolean distinguishing multiplication from accumulation. The Boolean gives the declared two-scalar-operation convention.

Source: `formalization/PacketCostModels.lean`.

Exact lines 68–73. Excerpt `bdlr_count`. SHA-256 in Section 15.

```
theorem bdlr_application_count {I : Type*} [Fintype I]
    (width : I → ℕ) (n r q : ℕ) :
    Fintype.card (BDLROperation I width n r q) =
      2 * q * ((∑ i, width i ^ 2) + 2 * n * r) := by
  simp [BDLROperation, pow_two]
  ring
```

Proof reading. The type `BDLROperation` is a sum of the diagonal-action, right-factor, and left-factor event types. `simp` evaluates cardinalities of finite products and sums, and `ring` rearranges the resulting polynomial. The count includes the accumulation into the diagonal result through the declared third kernel. It does not establish how many GPU instructions run.

The storage count concerns stored scalar values. Sparse index metadata and workspace are outside the corresponding models. The materialization theorem counts an explicit dense low-rank multiplication followed by one addition per output entry. Its quadratic remainder is exactly n^2. This is a count for that declared construction, rather than a lower bound for every conceivable implementation.

Source: `formalization/BandedCount.lean`.

Exact lines 73–77. Excerpt `band_count`. SHA-256 in Section 15.

```
theorem banded_block_count (p w : ℕ) (hw : w < p) :
    Fintype.card (BandedAddress p w) = p*(2*w+1)-w*(w+1) := by
  rw [bandCount_card]
  have he := bandCount_balance p w hw
  omega
```

Proof reading. The banded-count development counts the actual subtype of pairs in `Fin p × Fin p` whose distance is at most `w`. The premise `w < p` controls the two triangular boundary losses. The preceding recurrence and finite-sum lemmas establish the cardinality. This endpoint presents the closed formula. Replacing p by n/b additionally requires positive block width and exact divisibility.

FORMAL COMPANION XII

Patching Support and Hecke Packets

The formal companion to Chapter XII explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2618

Return to Chapter XII

```
  simp

theorem backward_forward (I : Ideal R) (x : Completion R / closedExtension I) :
    backward I (forward I x) = x := by
  obtain ⟨x, rfl⟩ := Ideal.Quotient.mk_surjective x
  exact backward_forwardMap I x

/-- Completion commutes with quotient by the closed extended ideal. -/
def equiv (I : Ideal R) :
    (Completion R / closedExtension I) ≃+* Completion (R / I) :=
  RingEquiv.ofRingHom (forward I) (backward I)
    (RingHom.ext (forward_backward I)) (RingHom.ext (backward_forward I))

theorem equiv_continuous (I : Ideal R) : Continuous (equiv I) := forward_continuous I

theorem equiv_symm_continuous (I : Ideal R) : Continuous (equiv I).symm :=
  backward_continuous I

@[simp] theorem equiv_mk_coe (I : Ideal R) (r : R) :
    equiv I (Ideal.Quotient.mk (closedExtension I) (r : Completion R)) =
      (Ideal.Quotient.mk I r : Completion (R / I)) := forwardMap_coe I r

/-- A bundled homeomorphism accompanying the ring equivalence. -/
def homeomorph (I : Ideal R) :
    (Completion R / closedExtension I) ≃ₜ Completion (R / I) where
  toEquiv := (equiv I).toEquiv
  continuous_toFun := equiv_continuous I
  continuous_invFun := equiv_symm_continuous I

end Countable
```

Applying this to the raw finite tensor quotient gives the displayed O-algebra equivalence of Claim 3.1. Its forward direction in the manuscript is the inverse of the construction direction. Both maps are continuous, and every factor identity is proved with the actual inclusions.

Source: `PaperXII/CompletedFiniteTensor.lean`, lines 91–133.

```
    CompletedTensor (fun i ↦ R i / J i) mO
      (fun i ↦ (m i).map (Ideal.Quotient.mk (J i))) ≃ₐ[O]
      (CompletedTensor R mO m / closedConditions R mO m J) :=
  (show (CompletedTensor R mO m / closedConditions R mO m J) ≃ₐ[O]
      CompletedTensor (fun i ↦ R i / J i) mO
        (fun i ↦ (m i).map (Ideal.Quotient.mk (J i))) from
    { __ := completedQuotientEquiv R mO m J
      commutes' r := by
        change completedQuotientEquiv R mO m J
          (Ideal.Quotient.mk _ (tensorEmbed R mO m (algebraMap O (Raw (O := O) R) r))) = _
        rw [completedQuotientEquiv_mk_embed, (quotientMap R J).commutes]
        rfl }).symm

/-- Both maps of the completed tensor quotient equivalence are continuous. -/
theorem completed_tensor_quotient_continuous :
    Continuous (completed_tensor_quotient R mO m J) ∧
      Continuous (completed_tensor_quotient R mO m J).symm :=
  ⟨completedQuotientEquiv_symm_continuous R mO m J,
    completedQuotientEquiv_continuous R mO m J⟩

/-- The isomorphism respects every actual canonical factor map. -/
@[simp] theorem completed_tensor_quotient_factor (i : ι) (x : R i) :
    completed_tensor_quotient R mO m J
      (completedInclude (fun i ↦ R i / J i) mO
        (fun i ↦ (m i).map (Ideal.Quotient.mk (J i))) i (Ideal.Quotient.mk (J i) x)) =
      Ideal.Quotient.mk _ (completedInclude R mO m i x) := by
```

Glossary

Definitions and reading references for the recurring language of the twelve chapters. Local meanings and hypotheses remain those of the cited mathematical treatment.

BROWSE ALPHABETICALLY

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R	2659	S	2661	T	2663
V	2664	W	2664	Z	2664

A

Automorphic Hecke datum

A Hecke algebra with a finite automorphic module or perfect complex, declared unramified Hecke polynomials, and a characteristic zero locus where classicality and an attached Galois representation are supplied. These realization inputs are part of the datum.

Chapter XII, Definition 9.1

See also: Hecke packet, Residual automorphic occupancy

B

Block diagonal plus low rank operator

An operator represented as a block diagonal part plus one thin factor times the adjoint of another. This separates independent packet work from a coupling term of rank at most the factor width. The abbreviation is BDLR.

Chapter VIII, Definition 3.3

See also: Packet operator representation, Output contract

Bridge datum

In the restricted Galois to type comparison, a supplied symplectic identification of two envelopes, compatible stabilizer action, and a common central character. At the prime two, compatible quadratic refinements and a lift to the central extensions are also required.

Chapter XI, Definition 11.1

See also: Symplectic envelope, Quadratic refinement

C

Clifford cocycle class

The cohomology class of the scalar factors by which multiplication of projective multiplicity operators differs from the group law. In the one break setting, its commutator pairing determines the associated symplectic envelope.

Chapter XI, Definition 9.3

See also: Symplectic envelope, Bridge datum

Output contract

The mathematical object an algorithm is required to return, such as an explicit vector, matrix, factors, or an implicit operator with an application rule. Costs are comparable only after the represented object and required output are fixed.

Chapter VIII, Definition 2.7

See also: Packet operator representation, Block diagonal plus low rank operator

P

p-adic precision

A lower bound on the p-adic valuation in the declared integral coefficient ring. The Euler congruence statements assign infinite valuation to zero. Divisibility in the field alone does not express precision, and special finite conventions for a zero step are stated separately.

Chapter IX, Definition 3.4

See also: Congruential Euler coefficients, Forward difference, Hurwitz integral series

Packet

In the linear theory, a component selected by a declared projector, usually identified with its range and index. Analytic, signal and arithmetic realizations attach their own additional structure. The meaning of a packet is therefore read with its specified system.

Chapter I, Definition 7.1

See also: Packet system, Projector, Transfer block

Packet analysis and synthesis

Analysis sends an object to its tuple of projected packet components. Synthesis adds those components in the ambient space. For a finite complete packet system the maps are inverse isomorphisms. Infinite versions require a specified convergence mode and corresponding bounds.

Chapter I, Definition 7.4 and Theorem 7.5

See also: Packet system, Packet support, Refinement

Packet jet

A higher derivative restricted to specified input packet spaces and projected to a specified output packet. Summing over all addresses reconstructs the derivative exactly. The analytic statements require the stated bounded projectors and continuous Fréchet differentiability.

Chapter III, Definition 17.1

See also: Packet, Transfer block

V

Visibility weight

The first precision depth at which two distinct integral packet coordinates have different reductions, when such a depth exists. At the prime two, signs may share a residual reduction yet separate at a later depth. This weight measures separation of coordinates, not a projector denominator.

Chapter X, Definition 5.1

See also: Separation weight, Residual root, Seed zeta function

W

Weighted transfer series

A generating series for walks in a finite directed multigraph with positive integer edge weights and scalar multiplicities. It sums multiplicity products and endpoint weights at each total weight, including the empty walk. The positive weights make each coefficient well defined.

Chapter II, Definition 6.1

See also: Transfer graph, Complete multiplicative filtration

Z

Zero power sum

A sum of inverse powers of the reduced zeros of the cyclic analytic denominator, counted with multiplicity. For section order N at least two, these sums converge absolutely and agree with the normalized logarithmic coefficients through the genus-zero product.

Chapter IV, Definition 12.5 and Theorem 12.6

See also: Reduced denominator, Transfer tangent

Sources and references

Sources are grouped by chapter and treatment. Citation labels retain the numbering or short names used in the text. The accompanying files contain the editable material and supporting source records.

BY CHAPTER AND TREATMENT

Chapter I mathematical text 2666

Chapter II mathematical text 2668

Chapter III mathematical text 2670

Chapter IV mathematical text 2672

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Chapter XI mathematical text 2693

Arithmetic realization studies: direct construction 2695

Arithmetic realization studies: monomial routes and Clifford descent 2698

Arithmetic realization studies: conductors and native markings 2700

Chapter XII mathematical text 2702

Chapter I mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2667

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The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

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